

Precision-recall curves

Thresholds, the trade-off they set, and how to build, read and summarise the curve (running case: brick kilns).

THE QUESTION

A detector gives each image a kiln score. Where should the cut-off go, and how do we compare detectors without fixing one?

THE INTUITION

Lowering the threshold τ flags more images, so recall rises and precision usually falls. The PR curve plots (recall, precision) at every τ ; average precision sums it up in one number.

KEY TAKEAWAY

Pick τ by what a false alarm and a miss each cost. With rare positives, read PR and AP rather than accuracy or ROC; the PR floor is the positive rate.

scores $P(y = 1 | x)$ → threshold τ → confusion matrix → point (R, P) → sweep every τ → PR curve → AP · choose τ

1 Checking 10,000 images by hand takes 83 hours

inspector time at 30 s per image

check all 10,000 images by hand 83.3 h

aggressive model: 2,000 flags 16.7 h

cautious model: 100 flags 50 min

1.5% of images hold a kiln: 150 of 10,000. "No kiln" for every image scores 98.5% accuracy and finds none. Accuracy hides which kind of mistake is made.

Illegal brick kilns hurt air quality over thousands of square kilometres, so a model shortlists satellite images for an inspector. Models at 95%, 92% and 89% accuracy say nothing about which mistakes each one makes.

2 False alarms cost time, misses cost violations

model flags 100 images

80 TP
20 FP
precision
= 80/100
= 0.80

FP (type I): a false alarm, an inspector's wasted trip

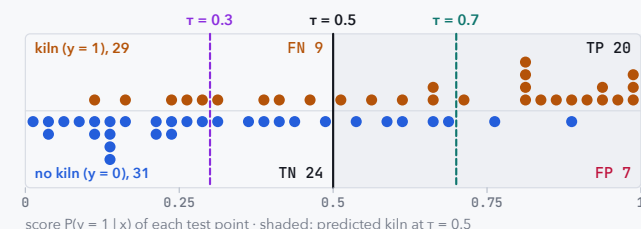
FN (type II): a missed kiln, a violation nobody sees

150 images really hold a kiln

80 TP
70 FN
recall
= 80/150
= 0.533

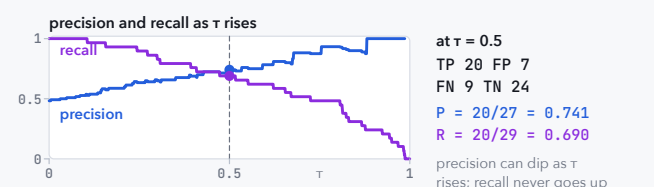
As in the conventions and metrics sheet: precision = $TP / (TP + FP)$, how often "kiln" is right; recall (sensitivity, TPR) = $TP / (TP + FN)$, the share of real kilns found. The slides' confusion matrix has actual classes in rows, as scikit-learn does; the conventions sheet puts predictions in rows.

3 A threshold τ turns scores into labels



Predict $\hat{y} = 1$ if $P(y = 1 | x) \geq \tau$ (default 0.5). A new τ changes the labels, not the model: a score of 0.62 is a kiln at $\tau = 0.5$ but not at 0.7. Data: the notebook's 100 two-feature blobs (spread 8); an unpenalised logistic regression trains on 40 and scores the other 60.

4 Raising τ trades recall for precision



A higher τ flags fewer images: fewer false alarms, more misses. Take a low τ (0.3) when a miss is costly, a high one (0.7) when a false alarm is. Recall never rises with τ ; precision usually does but can dip, from 0.933 at $\tau = 0.80$ to 0.900 at 0.85.

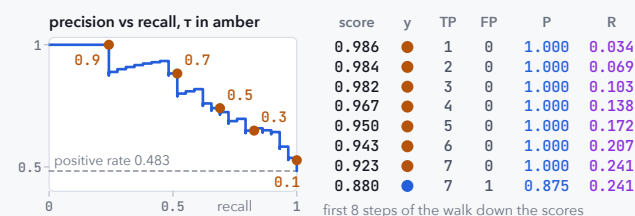
5 Each threshold gives one confusion matrix

τ	TP	FP	FN	TN	precision	recall	FPR	slides P, R
0.0	29	31	0	0	0.483	1.000	1.000	0.48 1.00
0.1	29	26	0	5	0.527	1.000	0.839	0.55 0.98
0.3	24	13	5	18	0.649	0.828	0.419	0.65 0.85
0.5	20	7	9	24	0.741	0.690	0.226	0.74 0.69
0.7	15	2	14	29	0.882	0.517	0.065	0.85 0.45
0.9	7	0	22	31	1.000	0.241	0.000	1.00 0.24

60 test points, 29 kilns · predict 1 when score $\geq \tau$ · red: a slide value that differs

Rebuilding the notebook's data reproduces all 21 rows of its sweep; the slides' table matches only at $\tau = 0, 0.5$ and 0.9. At $\tau = 0$ everything is flagged, so precision is the positive rate, $29/60 = 0.483$. At 0.9 every flag is right, but 76% of kilns are missed.

6 The PR curve joins every (recall, precision) point



Only the 60 distinct scores matter as thresholds. Sort them high to low and lower τ past one score at a time: a kiln adds a TP (recall and precision up), a non-kiln an FP (precision down, recall flat). The top seven scores are all kilns, so the curve starts at precision 1.

One number, one threshold, and the ROC view

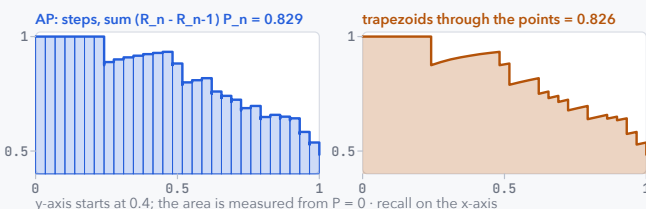
Baselines, average precision, F-scores, workloads, ROC and AUC, and why ROC flatters imbalanced data.

7 Up and right is better; the floor is the positive rate



A perfect detector keeps $P = 1$ up to $R = 1$. Random scores give precision near the positive rate at every recall: a flat no-skill line at positives / total (0.5 for balanced classes). If one curve lies above another everywhere, that model is better; if they cross, use the left part when precision matters and the right part when recall does.

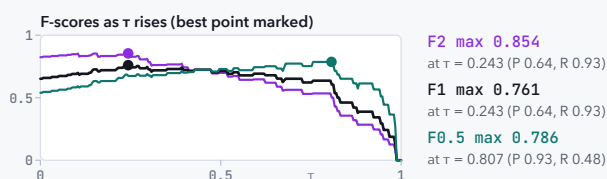
8 Average precision is a step sum



$$AP = \sum_n (R_n - R_{n-1}) P_n$$

weights each rise in recall by the precision at that threshold: 0.829, the notebook's "AUC-PR" from `average_precision_score`. Straight lines between points give 0.826, but no threshold reaches those in-between precisions, so report AP. It runs from 0 to 1; mAP, the mean AP over classes, scores object detectors.

9 F-scores pick a threshold



$F_1 = 2PR/(P + R) = 2 TP / (2 TP + FP + FN)$ is a harmonic mean, so it favours balance: $P = 0.8, R = 0.6$ gives 0.686, $P = R = 0.7$ gives 0.700. $F_\beta = (1 + \beta^2) PR / (\beta^2 P + R)$: $\beta > 1$ favours recall (F_2 : screening, kilns), $\beta < 1$ precision ($F_{0.5}$: spam, search). Best F_1 here is 0.761 at $\tau = 0.243$, against 0.714 at 0.5. Choose τ on validation data; these are test scores, for illustration.

10 A point on the curve is a workload

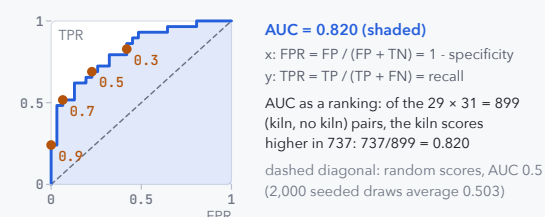
flagged images: TP (green) + FP (red)

A: $\tau = 0.7$ 170 +30 1.7 h, R 0.55
B: $\tau = 0.4$ 325 175 4.2 h, R 0.85
quiz, 2% faulty 180 120 300 flagged

30 s per flag. Quiz: 200 of 10,000 faulty, P 0.6, R 0.9, so $TP = 0.9 \times 200 = 180$ and flags = $180 / 0.6 = 300$.

Option A ($P = 0.85, R = 0.55$) sends 200 images and misses 45% of kilns; B ($P = 0.65, R = 0.85$) sends 500 and misses 15%. Compliance usually favours B: a miss costs more than inspector hours. Illustrative numbers (they imply 309 and 382 kilns).

11 ROC plots TPR against FPR



The receiver operating characteristic, first used on radar, traces (FPR, TPR) over all τ ; specificity is $TN / (TN + FP) = 1 - FPR$. Perfect hugs the top-left corner; below the diagonal is worse than random. $AUC = \int_0^1 TPR d(FPR)$ is the chance that a random positive outscore a random negative.

12 Rare positives: ROC holds, PR falls



Same 60 scores, each negative repeated. At $\tau = 0.5$ with $\times 50$: $FP = 350$, FPR still 0.226, precision 0.054; always saying "no kiln" scores 98.2% accuracy.

FPR divides by all negatives, so many negatives hide many false alarms: 100 among 950 non-kilns is FPR 0.105. Extra negatives leave ROC unchanged while precision and AP drop. Use PR when rare positives matter most (disease, fraud, kilns), ROC for balanced classes.

```
from sklearn.metrics import (
    precision_recall_curve, roc_curve,
    average_precision_score, roc_auc_score)
s = model.predict_proba(X_test)[: , 1]
P, R, thr = precision_recall_curve(y, s)
average_precision_score(y, s) # 0.829
fpr, tpr, _ = roc_curve(y, s)
roc_auc_score(y, s) # 0.820
```

lower τ : recall up, precision down · AP sums the PR curve · its floor is the positive rate · rare positives: PR over ROC

Next: the multi-layer perceptrons sheet. Source: lecture slides "Precision-recall curves and evaluation metrics" and the PR and ROC curves notebook (nipunbatra.github.io/ml-teaching); its data and logistic regression are rebuilt with NumPy, forest and SVM values quoted. Corrected: three rows of the slides' threshold table, two of their ROC table, and their model comparison table (the notebook ranks logistic regression first).